A Project Report

On

Modeling Antibody–Antigen Interactions Using AlphaFold3

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Certificate

This is to certify that the project report entitled "Modeling Antibody–Antigen Interactions Using AlphaFold3" submitted by Mr. Shubh Srivastava (ID No. 2022B4A70664H), Atharva Chaphekar (ID No. 2022B4A71815H), Ruchi Harge (ID No. 2022B4A70942H) and Rithika Balusu (ID No. 2022B1A70938H) in partial fulfillment of the requirements of the course MATH F376, Design Project Course, embodies the work done by them under my supervision and guidance.

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ABSTRACT

Predicting biomolecular structures and interactions is a central challenge in computational biology. Recent advances in machine learning, particularly the AlphaFold framework developed by DeepMind, have significantly improved the accuracy of protein structure prediction. AlphaFold3 extends this capability beyond individual proteins to model interactions between multiple biomolecules such as proteins, nucleic acids, and ligands.

This project explores the potential of using AlphaFold-based structural prediction to analyze antibody–antigen interactions and investigate whether engineered antibodies can hinder antigen binding to receptors. The study focuses on modeling interactions among vascular endothelial growth factors (VEGF), its receptor (VEGFR), and therapeutic antibodies such as Bevacizumab and Ranibizumab. Through multiple AlphaFold simulations executed on the BITS Hyderabad High Performance Computing (HPC) cluster, we analyze structural binding patterns and interaction strengths using metrics such as inter-chain predicted TM score (ipTM) and ranking confidence. The results demonstrate that AlphaFold can successfully model multi-molecular complexes and provide insights into how antibodies may disrupt receptor–ligand interactions.

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1. Introduction

Understanding biomolecular interactions is critical for drug design, immunology, and molecular biology. Antibody–antigen interactions play a vital role in immune defense and therapeutic interventions against diseases such as cancer and viral infections.

Recent developments in deep learning have dramatically improved the ability to predict protein structures from amino acid sequences. Among these developments, AlphaFold, developed by DeepMind, represents one of the most significant breakthroughs in computational biology.

The latest generation of this framework, AlphaFold3, expands the scope of structural prediction to include complex biomolecular assemblies involving proteins, DNA, RNA, small molecules, and ligands. This capability allows researchers to investigate molecular mechanisms such as receptor binding, antibody neutralization, and drug interactions with unprecedented accuracy.

This project investigates whether AlphaFold can be used to simulate antibody-mediated inhibition of antigen binding. Specifically, we attempt to model interactions between VEGFA (vascular endothelial growth factor A), VEGFR (vascular endothelial growth factor receptor), and antibodies such as Bevacizumab and Ranibizumab. By modeling these complexes, we aim to analyze whether antibody binding interferes with the VEGFA–VEGFR interaction.

2. Background: AlphaFold3

AlphaFold3 represents the latest generation of the AlphaFold framework developed by Google DeepMind. While AlphaFold2 primarily focused on predicting the three-dimensional structures of individual proteins, AlphaFold3 significantly expands this capability to include complex biomolecular interactions.

The model can simultaneously predict structures involving protein–protein interactions, protein–DNA interactions, protein–RNA interactions, and protein–ligand complexes.

The key innovation in AlphaFold3 is the integration of diffusion-based generative modeling with attention-based neural architectures.

Traditional protein structure prediction models rely on deterministic refinement approaches. In contrast, AlphaFold3 generates molecular structures by starting from noisy atomic configurations and gradually refining them into physically plausible biomolecular assemblies using a diffusion process.

This generative modeling approach allows AlphaFold3 to represent flexible molecular conformations and predict multi-molecular complexes. Due to these improvements, AlphaFold3 has demonstrated strong performance in predicting protein complexes and ligand binding interactions, making it highly valuable in fields such as drug discovery, structural biology, and molecular immunology.

3. Project Overview

The goal of this project is to explore whether AlphaFold-based structural modeling can help analyze the effects of antibodies on receptor–ligand interactions.

More specifically, we investigate whether the presence of an antibody can hinder the interaction between an antigen and its receptor.

This concept is particularly relevant in biological scenarios such as viral infection, where antibodies may bind to viral proteins and prevent them from attaching to host receptors.

The main objectives of the project are:

1. Model interactions between antibodies, antigens, and receptors using AlphaFold.
2. Determine whether antibody binding interferes with receptor–antigen interactions.
3. Estimate the minimum distance from the binding site where an antibody can effectively hinder antigen binding.
4. Analyze structural interactions using molecular visualization tools.

The project workflow involves generating input sequence files in JSON format, running AlphaFold simulations on the HPC cluster, analyzing predicted complexes using structural metrics, and measuring molecular distances using visualization software.

For structural analysis, ChimeraX is used to measure distances between interacting residues and identify binding interfaces.

4. Computational Infrastructure

All AlphaFold simulations were executed on the BITS Hyderabad High Performance Computing (HPC) cluster. Due to the computationally intensive nature of AlphaFold3, the use of HPC resources with GPUs is essential.

The workflow includes preparing sequence input files in JSON format, submitting jobs to the HPC scheduler, running AlphaFold using GPU nodes, and collecting and analyzing output structures.

5. Running AlphaFold on the HPC Cluster Using SLURM

The BITS Hyderabad HPC cluster uses the SLURM workload manager to schedule computational jobs.

A SLURM job script is required to request computational resources such as GPUs, CPUs, memory, and runtime.

Example SLURM Script:

```
#!/bin/bash
#SBATCH --job-name=alphafold_run
#SBATCH --output=alphafold.out
#SBATCH --error=alphafold.err
#SBATCH --partition=gpu
#SBATCH --gres=gpu:1
#SBATCH --cpus-per-task=8
#SBATCH --mem=64G
#SBATCH --time=48:00:00

module load alphafold

INPUT_JSON=input.json
OUTPUT_DIR=output/

alphafold3 --input_json $INPUT_JSON --output_dir $OUTPUT_DIR
```

Explanation of SLURM Parameters:

--job-name specifies the name of the job submitted to the scheduler.

--output specifies the file where standard output logs are written.

--error specifies the file where error logs are written.

--partition selects the GPU partition for execution.

--gres=gpu:1 requests one GPU for the job.

--cpus-per-task specifies the number of CPU cores allocated.

--mem specifies the amount of memory allocated.

--time specifies the maximum allowed runtime.

After creating the SLURM script, the job can be submitted using the command:

sbatch alphafold.slurm

Once submitted, the HPC scheduler assigns the job to an available GPU node where AlphaFold performs the structural prediction. The output directory contains predicted structures and confidence metrics such as pTM, ipTM, and ranking scores.

6. Experimental Runs and Observations

Run 1: VEGFR–VEGFA Complex (Initial Validation)

The first experiment aimed to verify whether AlphaFold3 was correctly configured and running on the HPC system.

The VEGFR–VEGFA complex was selected as a test case. A JSON file containing one sequence each for VEGFR and VEGFA was prepared and submitted.

However, the predicted structure differed significantly from the structure available in the Protein Data Bank. Further analysis revealed that this discrepancy occurred because the correct biological assembly requires two chains of each protein, as VEGFR and VEGFA form dimerized complexes.

Run 2: Corrected VEGFR–VEGFA Complex

In the second run, the input JSON file was modified to include two chains of VEGFR and two chains of VEGFA.

The resulting structure closely resembled the experimentally known structure, although the ends of some chains appeared extended. This occurred because terminal regions that become detached during complex formation were included in the input sequences.

Key Metrics:

Fraction Disordered: 0.14
Ranking Score: 0.89
ipTM Score: 0.89

These values indicate a highly confident structural prediction.

Run 3: Antibody–VEGFA–VEGFR Interactions

Three structural models were generated in this run.

Model 1: Bevacizumab Fab Mutant with VEGFA

Chains used:

Two heavy chains of Bevacizumab
Two light chains of Bevacizumab
Two VEGFA chains

The chain termini were trimmed to avoid unrealistic extensions.

Key results showed strong binding within antibody chains and moderate interaction with VEGFA.

Metrics:

ipTM: 0.35
Ranking score: 0.38

Model 2: Bevacizumab with VEGFA and VEGFR

This model included Bevacizumab Fab chains, VEGFA chains, and VEGFR domains D2 and D3.

The purpose of this model was to determine whether the presence of Bevacizumab interferes with VEGFA–VEGFR binding.

Results indicated that VEGFA had a stronger affinity for VEGFR compared to Bevacizumab.

Metrics:

ipTM: 0.37
Ranking score: 0.39
Chain pTM values ranged between 0.74 and 0.78.

The reduced ipTM compared to Run 2 suggests that the presence of Bevacizumab significantly affects the VEGFA–VEGFR interaction.

Model 3: VEGFA with VEGFR Domains D1 and D2

This model was created to compare interaction strength between different receptor domains.

Chains included two VEGFR chains containing domains D1 and D2 and two VEGFA chains.

Metrics:

ipTM: 0.78
Ranking score: 0.85

Although slightly lower than the D2–D3 configuration, the interaction remained strong. This suggests that domains D2 and D3 are the most significant contributors to binding, while domains D1 and D2 produce slightly weaker interactions.

Run 4: Ranibizumab Models

Two additional models were generated using the antibody Ranibizumab.

Model 1: included two VEGFA chains along with two heavy and two light antibody chains.

Model 2: included two VEGFA chains, two VEGFR chains, and the heavy and light chains of Ranibizumab.

These models were created to further analyze how therapeutic antibodies influence VEGFA binding to its receptor.

7. Structural Analysis Using ChimeraX

The predicted structures were analyzed using ChimeraX, a molecular visualization tool.

Using **ChimeraX,** distances between interacting residues and closest atomic contacts between molecules were measured. This allowed identification of binding interfaces between chains.

The distance measurement tool in ChimeraX helps determine the closest atoms involved in binding interactions, allowing estimation of the spatial threshold at which antibodies may hinder receptor–ligand binding.

8. Summary

The experiments demonstrate that AlphaFold can successfully model multi-molecular complexes and provide insights into molecular binding interactions.

Accurate prediction requires correct biological assembly information such as chain multiplicity. Structural confidence metrics including ipTM and ranking score provide useful indicators of binding reliability.

The presence of antibodies such as Bevacizumab significantly alters VEGFA–VEGFR binding interactions, suggesting that AlphaFold can be used as a computational tool to study antibody-mediated inhibition mechanisms.

9. Future Work

Future work will focus on quantifying molecular distances between antibody and receptor binding sites and performing mutation analysis on antibodies.

We aim to investigate whether mutated antibodies can effectively block receptor–ligand interactions and extend this analysis to virus–receptor–antibody systems.

Such studies may contribute to the development of antibody-based therapeutic strategies capable of counteracting rapidly mutating viral proteins.

10. References

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